ML/DL Problem Solving Worksheet

A structured guide from data to deployment

Overview

This worksheet guides you through the essential steps for solving a machine learning or deep learning problem. Fill out each section thoughtfully to design a robust solution.

1. Problem Statement

- What is the problem type? (e.g., classification, regression, segmentation)
- What are the goals and success criteria?
- What is the expected output?
- What constraints exist (real-time, hardware, etc.)?

Your Response

2. Data Understanding

- What is the nature of the dataset? (images, text, tabular, time series)
- What are the data sources?
- Is the data labeled, balanced, complete?
- Any potential biases or ethical issues?

Your Response

3. Data Preprocessing

- What preprocessing steps are required (normalization, encoding)?
- Will you perform feature engineering?
- Will you augment the data or reduce dimensionality?

Your Response

4. Model Selection

- What model(s) are suitable?
- Will you use a pre-trained model or build from scratch?
- What is the reasoning behind the choice?

Your Response

5. Cost Function

- Which loss function will you use?
- Is a custom loss function needed?
- Is the loss aligned with the evaluation metric?

Your Response

6. Optimizer

- Which optimizer will you use (Adam, SGD, RMSprop)?
- What learning rate strategy?
- Will you apply regularization or gradient clipping?

Your Response

7. Training Strategy

- What is the train/val/test split?
- What batch size, epochs, callbacks will you use?
- Are you using early stopping, checkpoints, or LR scheduling?

Your Response

8. Evaluation Metrics

- Which metrics will you track (Accuracy, F1, MSE)?
- How will you validate the model?
- Will you use cross-validation or test on edge cases?

Your Response

9. Deployment Plan

- Where will the model be deployed (cloud, edge)?
- What is the inference latency/size target?
- Will you use containers, APIs, or embedded deployment?
- How will you monitor the model post-deployment?

Your Response

10. Risks and Constraints

- What are the key risks (bias, overfitting, data drift)?
- What constraints are you working under (time, hardware, budget)?
- What mitigation strategies are in place?

Your Response

11. Notes and Insights

- Any unexpected challenges or discoveries?
- Lessons learned or ideas for improvement?

Your Response